

Aptura Tech Solutions

Data Science Internship — Batch 02

Week 01 Task: Exploratory Data Analysis Report

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Dataset: Diamonds (seaborn built-in dataset, 53,940 records, 10 features)

Target Variable: price (USD)

1. Objective

This report presents a complete exploratory data analysis of the Diamonds dataset. It covers data cleaning, missing value handling, feature analysis, visualization, and correlation analysis, and answers the two assignment questions regarding feature influence and scalability of the workflow to larger datasets.

2. Data Cleaning Summary

Step	Result
Original rows	53940
Duplicate rows removed	146
Zero-value dimensions (x, y, z) treated as missing	flagged and imputed with median
Outlier rows removed (carat, IQR method)	38
Final cleaned rows	53756

3. Missing Value Handling

The raw dataset contained no null values in its original form, but the x, y, and z dimension fields contained physically invalid zero entries. These were treated as missing values and imputed using the column median, since diamond dimensions are tightly correlated with carat and a median imputation preserves the overall distribution without introducing bias from extreme values.

4. Visualizations

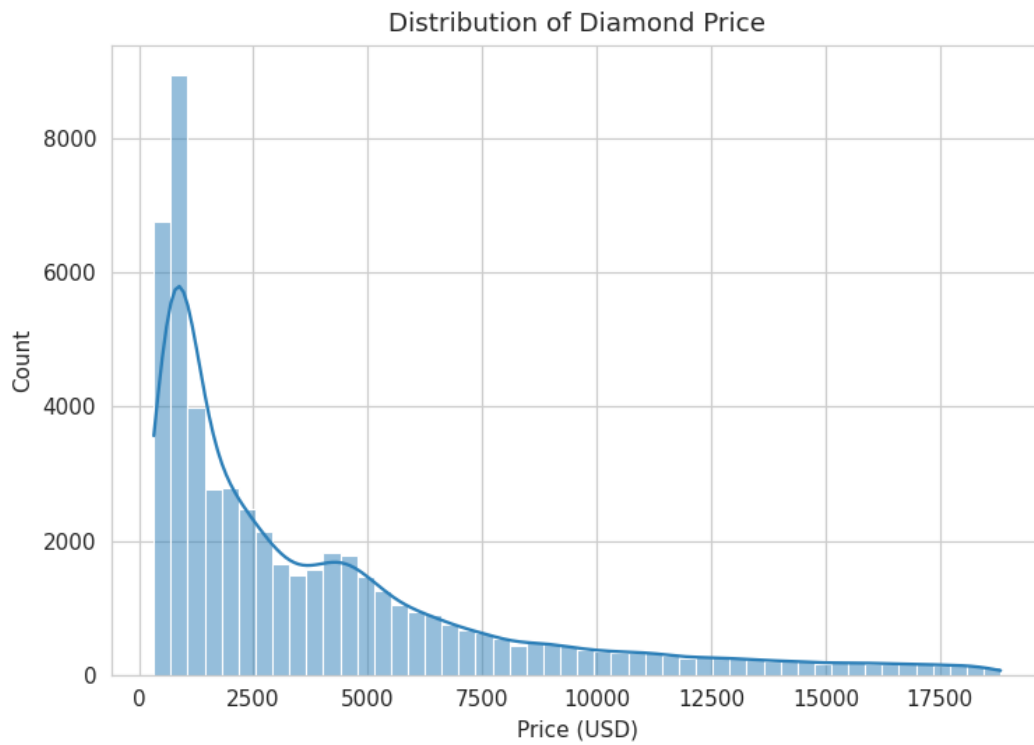


Figure 1: Distribution of diamond price. The distribution is right-skewed, with the majority of diamonds priced under \$5,000.

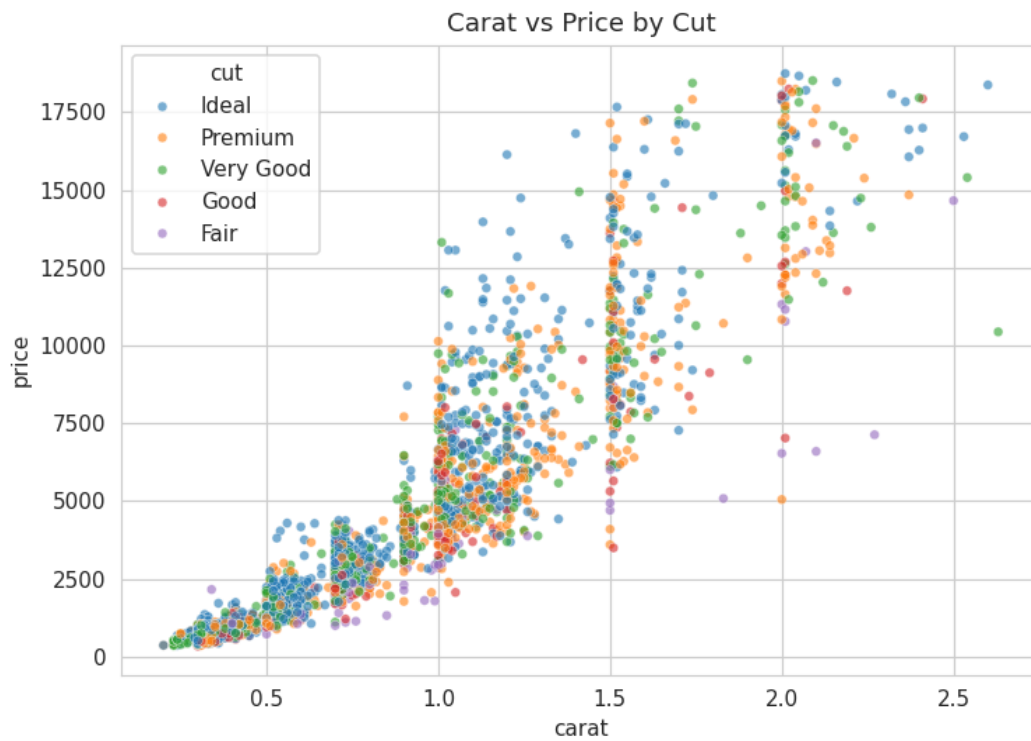


Figure 2: Carat vs price, colored by cut. A strong positive, near-exponential relationship is visible.

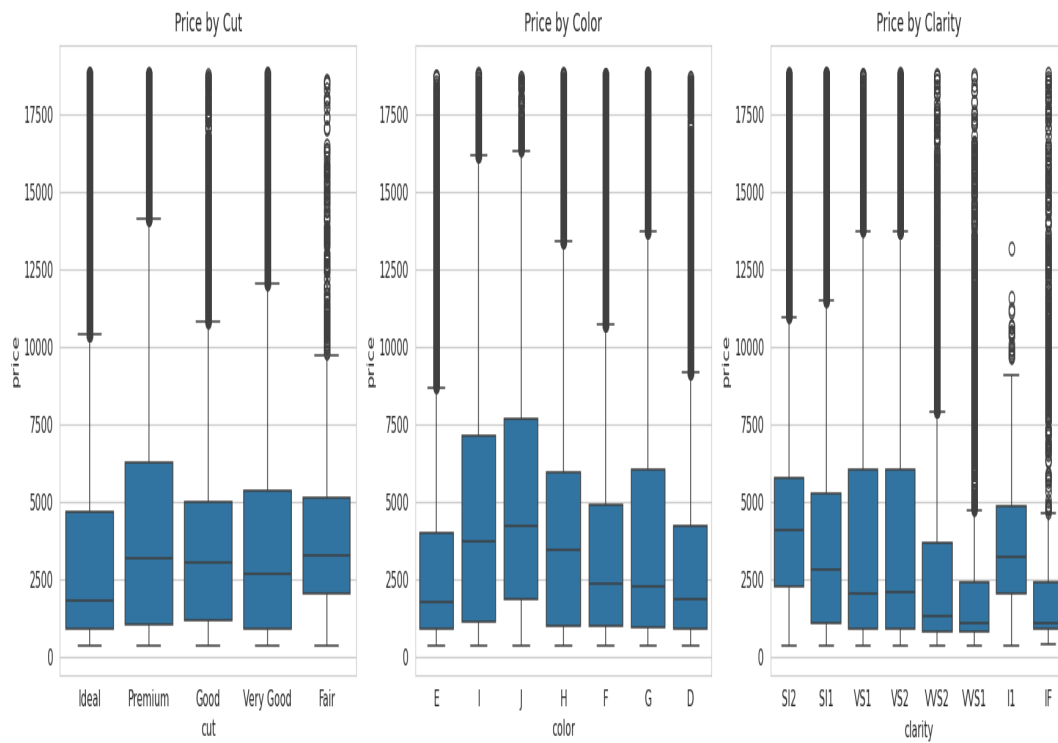


Figure 3: Price distribution across cut, color, and clarity categories.

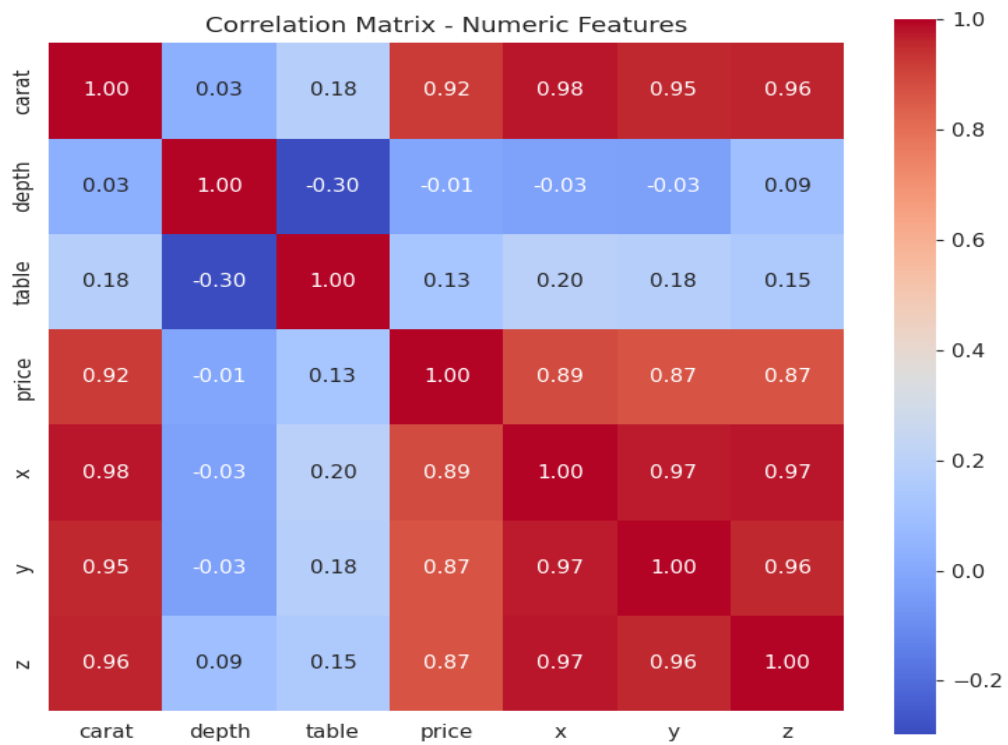


Figure 4: Correlation matrix of numeric features against price.

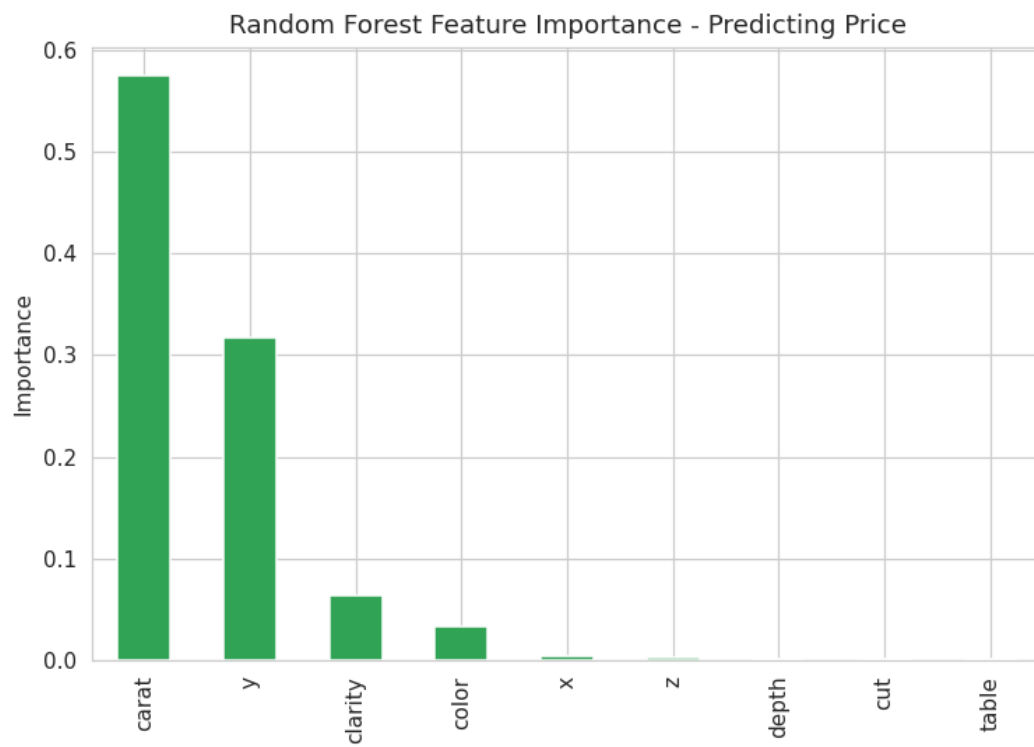


Figure 5: Random Forest feature importance ranking for predicting price.

5. Correlation Analysis

Feature	Correlation with Price
carat	0.923
x	0.887
y	0.867
z	0.867
table	0.126
depth	-0.012

6. Question 1 — Three Most Influential Features

A Random Forest Regressor (200 trees, max depth 12) was trained to predict price, achieving an R-squared of 0.9822 on a held-out test set. Feature importance rankings, cross-validated against the Pearson correlation matrix, identify the following as the three most influential features:

Rank	Feature	RF Importance	Correlation with Price
1	carat	0.574	0.923
2	y	0.3177	0.867
3	clarity	0.064	n/a

Carat is the dominant driver of price, both by correlation (0.92) and Random Forest importance (0.57), consistent with standard diamond market pricing. **The y dimension** (width in mm) ranks second, largely because it is mathematically linked to carat and volume. **Clarity** ranks third, contributing the strongest influence among the non-size-related features, since it directly reflects a diamond's visual quality grade.

7. Question 2 — Scaling to 10 Million Rows

Key challenges at scale:

- Memory limitations: a 10 million row dataset cannot be reliably held in RAM with standard pandas on a typical machine.
- Slow computation: correlation, groupby, and pairplot operations do not scale efficiently on the full in-memory dataset.
- Visualization overload: plotting millions of points directly produces unreadable, overplotted charts.
- I/O bottlenecks: repeated reads/writes of a single large CSV file become a major bottleneck.
- Data quality at scale: missing values, duplicates, and outliers become harder to detect manually and require automated routines.

Redesigned workflow:

- Convert storage to a columnar format such as Parquet for faster, compressed reads and writes.

- Use chunked reads (pandas chunksize) or out-of-core tools such as Dask, Polars, or Vaex for larger-than-memory processing.
- For scale beyond a single machine, move to a distributed engine such as PySpark.
- Perform exploratory analysis on a representative sample (1-5 percent), then validate findings against the full dataset.
- Use aggregated visualizations (hexbin, 2D histograms, downsampled scatter plots) instead of plotting every raw point.
- Downcast numeric types and convert repeated strings to the category dtype to reduce memory footprint.
- Automate the workflow into a checkpointed pipeline rather than an interactive notebook, so long jobs can resume after failure.

8. Conclusion

The cleaned Diamonds dataset confirms that price is overwhelmingly explained by physical size, with carat as the single strongest driver, followed by correlated dimension fields and clarity grade. The Random Forest model explains approximately 98 percent of price variance on unseen data. Scaling this workflow to 10 million records would require a shift from in-memory pandas processing toward columnar storage, sampling strategies, and distributed or out-of-core computation frameworks.